Teach2Learn: An AI-Driven Personalized Tutor Recommendation System

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CERTIFICATE

This is to certify that this internship report "**An AI-Driven Personalized Tutor Recommendation System**" is the Bonafide work of "**Mogalapalli Mounika (Y23CD122), Kasireddy Bhavya Sri(Y23CD090)**" who has carried out the work under my supervision and submitted in partial fulfilment for the award of **Project-I** (CD-363) during the year 2025-2026.

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ABSTRACT

Teach2Learn: An AI-Driven Personalized Tutor is a mini project that focuses on developing an intelligent learning platform that connects students with suitable tutors using Artificial Intelligence techniques. The system aims to enhance the learning experience by providing personalized tutor recommendations based on student academic details, learning preferences, and performance data.

The proposed system collects and processes data related to students such as academic level, subject interests, and learning requirements, along with tutor information including expertise, experience, availability, and ratings. Machine Learning algorithms are used to analyze this data and generate accurate tutor recommendations. Deep Learning models further improve the recommendation accuracy by learning from historical interactions and feedback provided after learning sessions.

The application is developed using Python and Flask for the backend, with HTML, CSS, and JavaScript for the frontend, and MySQL/SQLite for data storage. The system supports three types of users—students, tutors, and administrators—each with specific roles and functionalities. Feedback-driven learning enables continuous improvement in recommendation quality.

This project demonstrates the effective use of AI, Machine Learning, and Deep Learning in the education domain, offering a scalable and user-friendly solution for personalized learning. Teach2Learn provides a strong foundation for future enhancements such as real-time analytics, mobile integration, and advanced adaptive learning mechanisms.

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CHAPTER–1:
Introduction

1.1 Introduction of the Project

Education plays a vital role in shaping individuals and society. With the rapid advancement of technology, traditional teaching and learning methodologies are gradually being transformed into digital and intelligent learning systems. However, even with the availability of online learning platforms, students often face challenges in finding the right tutor who can understand their learning needs, academic level, and preferred learning style. Each student learns differently, and a one-size-fits-all approach is not always effective.

Teach2Learn: An AI-Driven Personalized Tutor is a smart learning platform designed to address these challenges by leveraging Artificial Intelligence (AI), Machine Learning (ML), and Deep Learning (DL) technologies. The primary objective of this project is to provide personalized tutor recommendations by analyzing student profiles, tutor expertise, and historical feedback data. The system ensures that students are matched with tutors who best suit their academic requirements, thereby improving learning efficiency and outcomes.

The project introduces an intelligent recommendation mechanism that automates the tutor selection process. By adopting a data-driven approach, Teach2Learn minimizes human bias, reduces time consumption, and enhances the overall learning experience. This system also benefits tutors by connecting them with suitable learners and increasing their reach through an organized digital platform.

1.2 Existing System

In the existing education system, the process of finding tutors is largely manual and inefficient. Students typically rely on coaching centers, personal referrals, advertisements, or generic online tutor listings. These methods do not provide intelligent matching between students and tutors, often resulting in poor compatibility and unsatisfactory learning experiences.

Most existing online platforms offer basic filtering options such as subject, location, or availability. However, they lack advanced personalization features and do not consider factors like student learning preferences, performance history, or tutor effectiveness. Additionally, feedback collected from students is rarely utilized to improve future tutor recommendations.

As a result, students spend a significant amount of time searching for suitable tutors, and tutors may not reach learners who truly match their expertise. The absence of automation, personalization, and continuous improvement makes the existing system inefficient and less effective.

1.3 Proposed System

The proposed system, Teach2Learn, overcomes the limitations of the existing system by introducing an AI-driven personalized tutor recommendation framework. The system is developed as a web-based application that provides a centralized platform for students, tutors, and administrators.

Teach2Learn collects and analyzes various data points, including student academic details, learning preferences, subject interests, and tutor profiles such as expertise, experience, availability, and ratings. Machine Learning algorithms process this data to generate personalized tutor recommendations, while Deep Learning models enhance accuracy by identifying complex patterns in user behaviour and feedback.

The system continuously improves its recommendations through feedback provided after learning sessions. This feedback-driven learning approach ensures adaptability and relevance over time. The proposed system reduces manual effort, improves tutor–student compatibility, and delivers a smarter and more efficient learning environment.

1.4 Potential Users

The Teach2Learn system is designed to support multiple user roles, each with specific responsibilities and functionalities. The primary users of the application include students, tutors, and administrators.

Students use the platform to register, create academic profiles, and receive personalized tutor recommendations based on their learning needs. They can also provide feedback after sessions, which helps improve the system's recommendation accuracy.

Tutors register on the platform to offer teaching services. They can manage their profiles by specifying expertise, qualifications, experience, and availability. Tutors receive requests from students who match their profiles and can build credibility through ratings and feedback.

Administrators are responsible for managing the overall system. They monitor user activity, maintain data integrity, manage user accounts, and ensure the smooth functioning and security of the platform.

1.5 Unique Features of the System

Teach2Learn offers several unique features that distinguish it from traditional tutor-finding platforms. The system is designed to provide an intelligent, personalized, and scalable learning experience by leveraging Artificial Intelligence technologies. Unlike conventional systems that rely on static filters, Teach2Learn dynamically analyzes user data and adapts its recommendations based on real-time interactions and feedback.

The system incorporates an AI-based recommendation engine that matches students with tutors according to individual academic needs, learning preferences, and historical performance. This data-driven approach ensures higher accuracy in tutor selection and improves overall learning outcomes. The continuous learning mechanism enables the system to evolve over time, making recommendations more relevant and effective.

Key unique features of the Teach2Learn system include:

- **AI-Based Personalized Tutor Recommendation:** Uses Machine Learning and Deep Learning algorithms to suggest tutors tailored to individual student requirements.
- **Feedback-Driven Improvement:** Continuously learns from student feedback to enhance recommendation accuracy and system performance.
- **User-Friendly Interface:** Designed for easy navigation and accessibility, suitable for users with varying levels of technical expertise.
- **Secure Authentication and Data Management:** Ensures protection of user information through secure login mechanisms and structured database storage.
- **Scalable System Architecture:** Built using modern web technologies that support system expansion and performance optimization.
- **Support for Peer-to-Peer Learning:** Enables direct interaction between students and tutors, fostering collaborative learning.
- **Future Enhancement Flexibility:** Supports future upgrades such as mobile application integration, real-time analytics, and advanced adaptive learning features.

These unique features make Teach2Learn a reliable, intelligent, and future-ready learning platform that enhances both teaching and learning experiences..

1.6 Demand for the Product

The demand for intelligent and personalized learning platforms has increased significantly due to the rise of online education, competitive academic environments, and remote learning needs. Students increasingly seek platforms that adapt to their individual learning styles and academic goals rather than offering generic solutions.

Teach2Learn addresses this growing demand by providing efficient tutor matching, reducing search time, and improving learning outcomes. Educational institutions, coaching centers, and independent learners can benefit from such a system. The integration of AI and data analytics makes Teach2Learn highly relevant in the modern digital education ecosystem.

1.7 Protection of Idea

The Teach2Learn system ensures protection of its core idea and user data through secure design practices. User authentication and authorization mechanisms prevent unauthorized access to sensitive information. Data is stored securely using structured database management techniques.

The originality of the project lies in its AI-driven personalized tutor recommendation approach and feedback-based learning model. The recommendation logic and system architecture are uniquely designed for educational applications, making the project suitable for academic evaluation and future commercialization.

CHAPTER–2:

Literature Survey / Related Work and SRS

2.1 Literature Survey

The rapid growth of online education platforms has led to increased research in personalized learning systems and intelligent recommendation mechanisms. Traditional e-learning systems primarily focus on content delivery rather than adapting to individual learner needs. To overcome this limitation, researchers have explored the use of Artificial Intelligence, Machine Learning, and Deep Learning techniques to personalize learning experiences.

Several studies highlight the effectiveness of recommendation systems in the education domain. Content-based filtering and collaborative filtering techniques have been widely used to recommend courses, learning materials, and tutors. These systems analyze user preferences, historical interactions, and ratings to generate recommendations. However, many existing systems rely on limited datasets and fail to adapt dynamically to changing user behavior.

Recent advancements in Deep Learning have enabled more accurate prediction models by capturing complex patterns in learner behavior. Neural networks and hybrid recommendation models have shown improved performance in educational data analysis. Feedback-driven learning systems have also been proposed to continuously improve recommendations based on learner satisfaction. These research findings form the foundation for the proposed Teach2Learn system.

2.1.1 Review Findings

From the literature survey, it is evident that intelligent recommendation systems significantly enhance learning efficiency when personalization is applied. Most successful systems utilize user data such as preferences, interaction history, and feedback. However, many existing solutions lack real-time adaptability, scalability, and integration of multiple user roles.

Another key observation is that feedback data is often underutilized. Systems that effectively incorporate feedback demonstrate higher accuracy and user satisfaction. These findings indicate the need for a robust, AI-driven tutor recommendation platform that continuously learns and improves—leading to the development of the Teach2Learn system.

2.1.2 Objectives of the System

The primary objective of the Teach2Learn system is to develop an intelligent platform that recommends suitable tutors to students based on their academic needs and preferences. The system aims to reduce the manual effort involved in tutor selection and enhance learning outcomes through personalization.

Additional objectives include providing a scalable and user-friendly platform, utilizing Machine Learning and Deep Learning techniques for accurate recommendations, enabling feedback-driven improvement, and supporting multiple user roles such as students, tutors, and administrators. The system also aims to ensure data security, reliability, and future extensibility.

2.2 Requirements Analysis

Requirements analysis is a critical phase that defines the functional and non-functional needs of the system. It ensures that the developed system meets user expectations and operates effectively under defined constraints. For Teach2Learn, requirements are identified based on user roles, system objectives, and technological feasibility.

2.2.1 Functional Requirements Analysis

Functional requirements describe the specific operations the system must perform. The Teach2Learn system must allow students and tutors to register and authenticate securely. Students should be able to create profiles, specify learning preferences, and receive tutor recommendations. Tutors should be able to manage their profiles, specify expertise, and receive student requests.

The system must support feedback submission after learning sessions and use this feedback to improve future recommendations. Administrators should have the ability to manage users, monitor data, and control system operations.

2.2.2 User Requirements

User requirements define the expectations of different users interacting with the system. Students require an easy-to-use interface, accurate tutor recommendations, and privacy of personal data. Tutors expect visibility, fair matching opportunities, and profile management features.

Administrators require tools to manage users, monitor performance, and ensure system security. The system must be accessible, responsive, and reliable for all user categories.

2.2.3 Non-Functional Requirements

Non-functional requirements focus on system quality attributes. Teach2Learn must provide high performance with quick response times for recommendation generation. The system should be scalable to support an increasing number of users and data volume.

Security is a critical requirement, ensuring protection of user data through authentication and authorization mechanisms. The system should also be reliable, user-friendly, and maintainable, allowing future enhancements without major redesign.

2.2.4 System Requirements

The system requires a web-based environment with a backend developed using Python and Flask. A relational database such as MySQL or SQLite is required for data storage. Machine Learning libraries like Scikit-learn and Deep Learning frameworks such as TensorFlow or Keras are required for model implementation.

The system should run on standard hardware with internet connectivity and be compatible with modern web browsers.

2.3 Modules Description

The Teach2Learn system is divided into multiple modules to ensure modularity and ease of maintenance. The Student Module handles registration, profile creation, tutor search, and feedback submission. The Tutor Module manages tutor registration, profile updates, expertise specification, and interaction with students.

The Recommendation Module is the core component that analyzes data and generates personalized tutor suggestions using AI models. The Admin Module manages system data, users, and overall monitoring. Each module interacts with the database to ensure seamless data flow.

2.4 Feasibility Study

Feasibility study evaluates the practicality of the proposed system in terms of technical, operational, and behavioral aspects. This analysis ensures that the project can be developed and implemented successfully within constraints.

2.4.1 Technical Feasibility

The project is technically feasible as it uses widely available technologies such as Python, Flask, and Machine Learning libraries. The required hardware and software resources are readily accessible, and the development team possesses the necessary technical skills.

2.4.2 Operational Feasibility

The system is operationally feasible as it is user-friendly and requires minimal training. Students, tutors, and administrators can easily interact with the system through an intuitive interface. The automation of tutor recommendation reduces manual workload.

2.4.3 Behavioural Feasibility

The system is behaviourally feasible as it aligns with user expectations and encourages adoption through personalized features. Users are likely to accept the system due to its ease of use and effectiveness.

2.5 Process Model Used

The Iterative Development Model is used for this project. This model allows incremental development, testing, and refinement of system components. Feedback from each iteration helps improve system functionality and performance, making it suitable for AI-based applications.

The development of the Teach2Learn: An AI-Driven Personalized Tutor system follows the Iterative Development Process Model, which is well-suited for projects involving Machine Learning and Artificial Intelligence components. This model emphasizes continuous development, testing, evaluation, and refinement of the system through multiple iterations rather than delivering the entire system in a single phase.

In this process model, the project is divided into smaller and manageable cycles, where each iteration focuses on implementing specific features such as user registration, profile management, tutor recommendation logic, and feedback analysis. After completing each iteration, the developed module is tested and evaluated to identify any issues or improvements required. This approach allows early detection of errors and ensures better quality at each stage of development.

The Iterative model is particularly beneficial for AI-driven systems because recommendation accuracy improves gradually as more data becomes available. Feedback collected from students after each learning session is analyzed and used to refine the Machine Learning and

Deep Learning models in subsequent iterations. This continuous learning and improvement process helps enhance system performance over time.

Additionally, the Iterative Development Model provides flexibility to incorporate new requirements or modifications based on user feedback without significantly affecting the overall system structure. This adaptability makes it an ideal choice for the Teach2Learn project, ensuring reliable development, improved accuracy, and efficient utilization of resources.

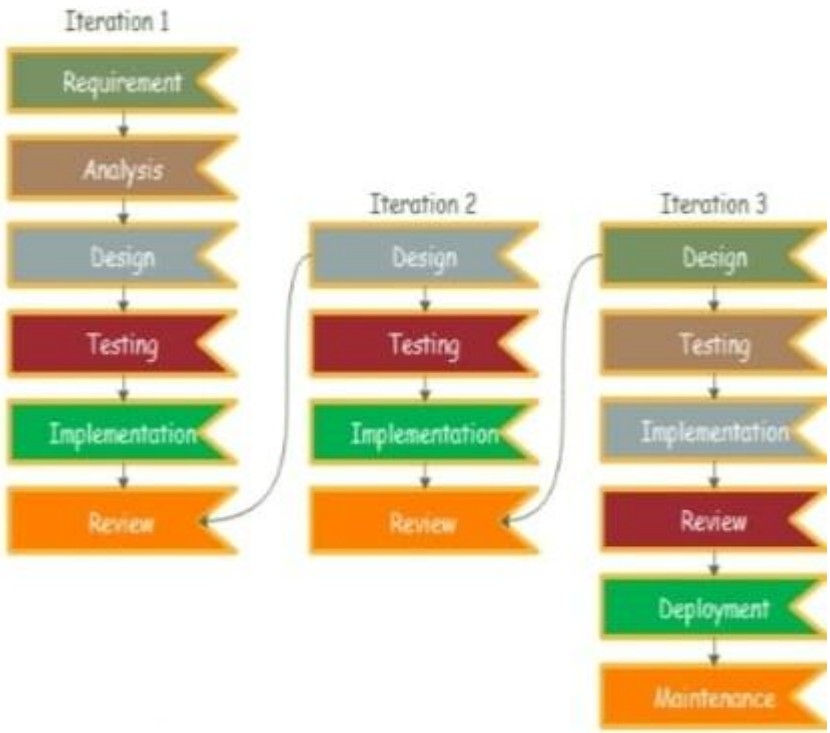

Fig 2.5 : Iterative Model

2.6 Hardware and Software Requirements

The Teach2Learn system requires specific hardware and software resources to ensure efficient development, execution, and scalability of the application. The requirements are as follows:

Hardware Requirements

- A personal computer or laptop for development and deployment
- Processor: Intel Core i3 or higher (or equivalent processor)
- Main Memory: Minimum 4 GB RAM (8 GB recommended for ML processing)
- Storage: Minimum 20 GB free disk space for application files and datasets
- Internet Connection: Stable internet connectivity for accessing the web application

- Input Devices: Keyboard and mouse for user interaction
- Output Devices: Monitor or display screen

Software Requirements

- Operating System: Windows 10 / Linux / macOS
- Programming Language: Python 3.x
- Backend Framework: Flask
- Frontend Technologies:
    - HTML for structure
    - CSS for styling
    - JavaScript for interactivity
- Database Management System: MySQL / SQLite
- Machine Learning Libraries:
    - Scikit-learn for ML algorithms
    - TensorFlow / Keras for Deep Learning models
- Development Tools:
    - Visual Studio Code / PyCharm for coding
    - Jupyter Notebook for model training and experimentation
- Web Browser: Google Chrome / Mozilla Firefox
- Version Control (Optional): Git

2.7 SRS Specification

The Software Requirements Specification (SRS) defines the overall functionality, constraints, and behavior of the Teach2Learn system. It includes system overview, functional and non-functional requirements, user roles, data flow, and system interfaces. The SRS ensures clarity between system design and implementation.

2.8 Financial Plan for the Development of Product (Optional)

Since Teach2Learn is a mini academic project, development costs are minimal. Open-source tools and libraries are used, reducing financial investment. Future commercialization may involve costs related to hosting, maintenance, and marketing.

2.9 Business Plan from Seeding to Commercialization (Optional)

Teach2Learn has strong potential for commercialization as an educational technology product. Initial deployment can target academic institutions and online learning communities. With increased adoption, advanced features and subscription-based services can be introduced.

2.9.1 Business Model Canvas

The business model includes value proposition through personalized learning, key users such as students and tutors, revenue streams through subscriptions or commissions, and key resources including AI models and platform infrastructure. This model supports sustainable growth.

CHAPTER–3:

Design Phase / System Design

3.1 Design Concepts and Constraints

The design phase of the Teach2Learn system focuses on converting the identified requirements into a structured and efficient system architecture. The primary design concept is to create an intelligent, scalable, and user-friendly platform that supports personalized tutor recommendations using Artificial Intelligence. Emphasis is placed on modularity to ensure that each functional component operates independently while remaining integrated within the overall system.

Another key design concept is scalability, as the system should be capable of handling an increasing number of students, tutors, and data records without significant performance degradation. The system is designed using a layered architecture that separates the frontend, backend, database, and AI models. This separation improves maintainability and allows future upgrades with minimal impact on existing components.

Usability is also a major design consideration. The interface is designed to be simple and intuitive so that users with minimal technical knowledge can easily navigate the platform. Clear workflows, minimal input complexity, and responsive interfaces ensure a positive user experience for students, tutors, and administrators.

Several constraints influence the design of the system. These include limited computational resources, availability of training data for AI models, and time constraints associated with academic project development. Additionally, the system must rely on open-source tools to minimize costs while ensuring data security, privacy, and compliance with ethical standards.

3.2 Design Diagram of the System

The system design of Teach2Learn follows a client–server architecture that supports efficient communication between users and backend services. The frontend acts as the client interface, where students, tutors, and administrators interact with the system through a web browser. This interface is responsible for capturing user input and displaying system outputs.

The backend server, developed using the Flask framework, acts as the core processing unit of the system. It handles user authentication, business logic, request processing, and coordination

between the database and AI models. All user requests from the frontend are routed to the backend for validation and processing.

The recommendation engine is integrated within the backend and is responsible for analyzing user data and generating personalized tutor recommendations. Machine Learning and Deep Learning models process academic data, tutor profiles, and feedback to produce accurate results. This engine plays a crucial role in the system's intelligence.

The database forms the foundation of the system by storing all persistent data, including user profiles, tutor details, feedback records, and recommendation history. The interaction between the frontend, backend, AI models, and database is clearly represented through system design diagrams, ensuring clarity and consistency in system operation.

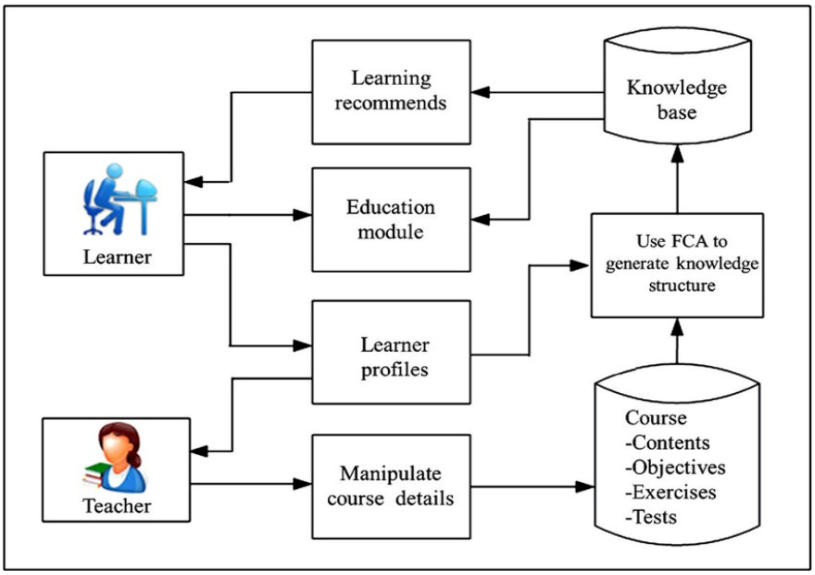

Fig 3.2: System Design Diagram

3.3 Conceptual Design

The conceptual design of Teach2Learn provides a high-level overview of the system and focuses on identifying key entities and their relationships. It abstracts technical complexities and presents a clear picture of how users and system components interact. This design phase helps stakeholders understand the system without delving into implementation details.

The primary entities in the system include Students, Tutors, Admin, Recommendation Engine, and Database. Students provide academic details and learning preferences, while tutors supply information regarding expertise, experience, and availability. These entities interact through the platform to facilitate effective learning connections.

The recommendation engine acts as the core conceptual component that processes user data and generates tutor suggestions. It uses stored information and feedback data to continuously improve recommendation quality. This feedback loop ensures adaptability and personalization within the system.

The conceptual design establishes clear boundaries between system components and defines data flow between entities. By representing relationships and interactions at a high level, this design ensures logical consistency, reduces ambiguity, and serves as a foundation for detailed logical and physical design phases.

3.4 Logical Design (Logical Tools / Logical Diagrams)

The logical design phase focuses on defining how the system functions internally by detailing data flow, process interactions, and system behavior. It bridges the gap between conceptual design and physical implementation by providing a structured representation of system logic.

Logical tools such as Data Flow Diagrams (DFDs) are used to illustrate how data moves between different modules of the Teach2Learn system. These diagrams depict processes such as user registration, profile management, tutor recommendation, and feedback analysis. They help visualize system workflows and identify data dependencies.

In the logical design, the student module interacts with the recommendation module to request suitable tutors. The tutor module updates profile and availability information, which is stored in the database. The admin module oversees user management and system monitoring. Each interaction is logically defined to ensure accuracy and consistency.

This design phase helps validate functional requirements and ensures that all system components communicate effectively. Logical design also simplifies debugging, testing, and future enhancements by clearly defining system behavior and data handling mechanisms.

3.5 Database Design

The database design of Teach2Learn is a critical component that ensures efficient data storage, retrieval, and management. A relational database model is adopted to maintain structured and organized data. This design supports data consistency and integrity across the system.

The database includes multiple tables such as Student, Tutor, Admin, Feedback, and Recommendation History. Each table is designed with appropriate attributes to store relevant

information. Primary keys uniquely identify records, while foreign keys establish relationships between tables.

Feedback data plays a crucial role in improving recommendation accuracy. Therefore, the database design ensures that feedback is properly linked to both students and tutors. This structured approach allows effective analysis of learning outcomes and system performance.

The database design also considers scalability and security. Proper indexing improves query performance, while access control mechanisms protect sensitive user data. This robust database structure supports the intelligent and data-driven nature of the Teach2Learn system.

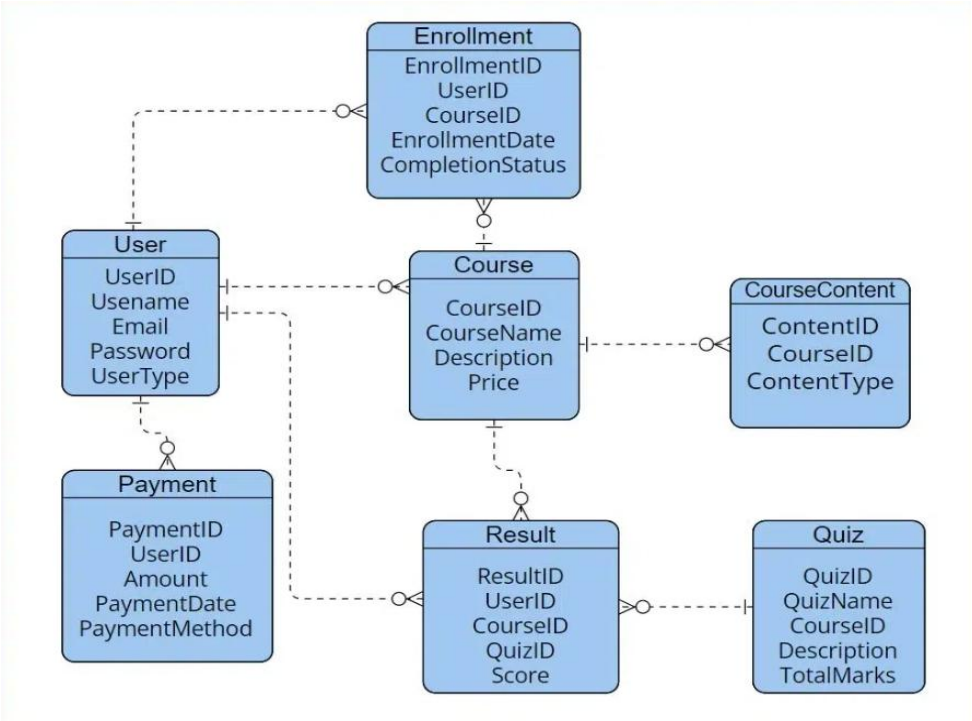

Fig 3.5: Database Design